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EXPLAINING PATIENT RETRIEVAL

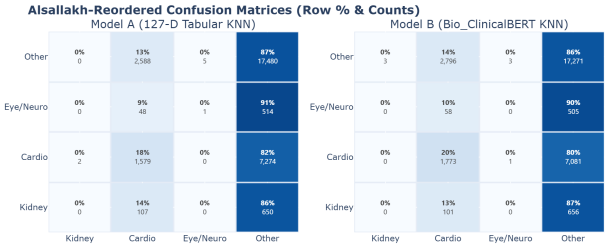


Team SHAP Shifters

Pair A (AI-Aided): Julian Bakan, Ahmad Hasan | **Pair B (Human-Only):** Alfiia Ziganshina, Aqeel Jajja
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1. Baseline Retrieval Paradox & Error Structure

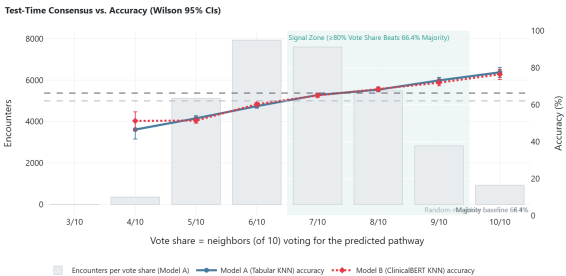


Diagnostic Findings

- **Cohort:** 30,248 repeat stays (16,773 patients).
- **Accuracy Paradox:** Both Model A (63.0%) and B (63.0%) trail the majority baseline (66.4%).
- **Class Collapse: 0.0% recall** on Kidney $(0/757)$ and Eye/Neuro $(0/563)$ admissions.
- **Disjoint Retrieval:** Models share only **22.4%** of retrieved neighbors $(2.2/10)$.

Key Takeaway: Both models trail the 66.4% majority baseline; flat $k = 10$ voting erases all rare complications.

2. Calibrating Trust: The $\geq 80\%$ Signal Zone

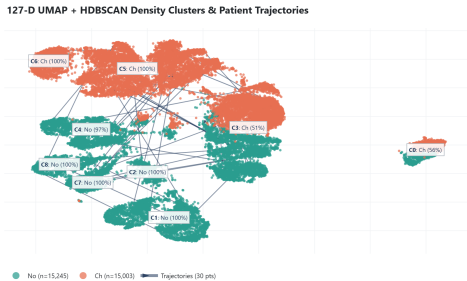


Operational Decision Rules

- **Noise Zone ($< 80\%$):** Covers **69.5%** of stays. Accuracy drops to **59.6%** — worse than random (62.1%).
- **Signal Zone ($\geq 80\%$):** Covers **30.5%** of stays. Accuracy jumps to **70.7%** (reaches **77.6%** when $10/10$).
- **Human AI Audit:** Caught circular AI metric using future labels; enforced test-time vote consensus.

Key Takeaway: Clinicians should abstain when neighbor consensus is $< 80\%$; unanimous votes yield $+11.2$ pp reliability gain.

3. 127-D Manifold Drivers: Workflow vs. Outcome



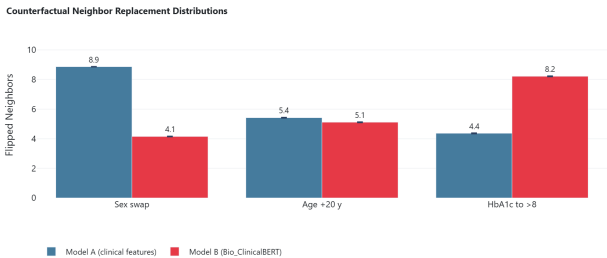
Quantitative Driver Ranking

Clinical Feature	Category	Driver Strength
`change`	Workflow	Cramér's $V = \mathbf{0.803}$
`diabetesMed`	Workflow	Cramér's $V = \mathbf{0.796}$
`max_glu_serum`	Lab Flag	Cramér's $V = \mathbf{0.518}$
`admission_source`	Administration	Cramér's $V = \mathbf{0.488}$
`insulin`	Medication	Cramér's $V = \mathbf{0.486}$
`num_lab_procs`	Utilization	Corr. Ratio $\eta = \mathbf{0.366}$
Next Pathway (V_{t+1})	Target	Cramér's $V = \mathbf{0.042}$ (NS)

Target Mixing: Future disease pathways are uniformly mixed across all UMAP clusters.

Key Takeaway: 127-D UMAP partitions patients by medication management protocol, not future disease trajectory ($V = 0.04$).

4. Perturbation Sensitivity: Mirror-Image Shortcuts

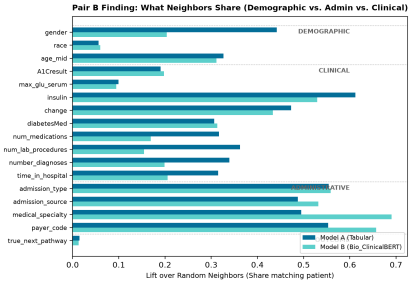


Mechanistic Vulnerabilities

- **Model A (Sex Shortcut):** Swapping sex flips **8.86** / **10** neighbors (**94.7%** sex homophily). Numeric fields were z-scaled, but one-hot flags unscaled: adds $\Delta\|x\|^2 = 2.0$.
- **Model B (Token Dilution):** Elevating HbA1c flips **8.22** / **10** neighbors (A: 4.36) because modifying string `'none'` $\rightarrow$ `'>8'` alters WordPiece tokens.

Key Takeaway: Mirror vulnerabilities: Model A is distorted by unscaled binary flags; Model B is fragile to token edits.

5. Neighborhood Geometry: Crowding & The Voting Mask

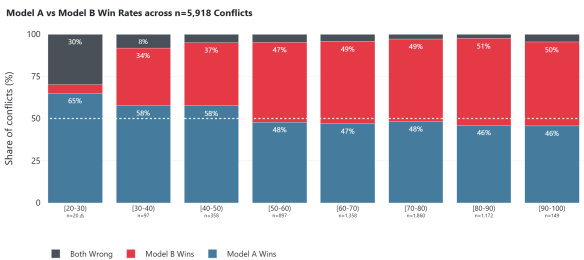


Pair B Human-Only Discoveries

- **Matching:** Model A matches on sex (+0.44 lift) and numeric labs/meds; Model B matches on administrative department.
- **Emergency Crowding:** 10,684 stays (35.3%) are ER admissions with missing specialty (’?’). Generic boilerplate packs them into ultra-dense space ($d_{10} = 0.0006$ vs. Model A 0.113).
- **Voting Mask:** In Eye/Neuro, Model A finds matching neighbors above chance, but flat voting silences them.

Key Takeaway: Model B fragility stems from 35% generic ER text crowding; majority voting masks real minority signals.

6. Conflict Specialization & Fairness Deconfounding



Clinical Specialization

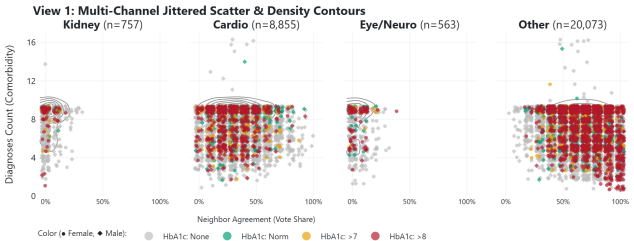
- **Younger (< 50y, $n = 482$):** Numeric lab counts favour Model A (**58.5%** vs. 34.9%).
- **Geriatric ($80+$y, $n = 1{,}321$):** Text captures multimorbidity in Model B (**51.2%** vs. 46.0%).

Fairness Deconfounding

- African American raw accuracy is 3.7 pp lower (60.0% vs. 63.7%).
- Base rate of uncomplicated stays is 3.8 pp lower (63.1% vs. 66.9%).
- Relative lift is identical (-3.1 pp vs. -3.3 pp).

Key Takeaway: Tabular KNN excels on acute youth readmissions; ClinicalBERT captures geriatric multimorbidity.

7. Coordinated Studio & Dynamic Cohort Profiler



Interactive Platform

- **Coordinated Views:** Lassoing patients in View 1 filters the UMAP (View 3), confusion matrix (View 2), and dossier flips (View 4).
- **Dynamic Profiler:** Evaluates clinical complexity shifts of any selected sub-cohort against the 100% hospital baseline.
- **Clinical Utility:** Audits individual neighbor breakdowns before admission decisions.

Key Takeaway: Interactive multi-view coordination allows clinicians to brush cohorts and audit local retrieval stability in real time.

8. Team Synthesis & Phase II Actionable Roadmap

AI-Aided vs. Human-Only Synthesis

Division of Strengths:

- **Pair A (AI):** Built interactive Dash studio, UMAP manifolds, and automated Cramér's V driver ranking.
- **Pair B (Human):** Uncovered ER text crowding bottleneck and majority vote suppression.

Human AI Auditing (3 Catches):

- Stripped AI-introduced target leakage proxy.
- Replaced circular calibration with test-time consensus.
- Deconfounded unnormalized racial disparity claim.

Phase II Actionable Milestones

1. Calibrated Thresholding ($\geq 80\%$)

Replace flat voting with consensus filtering to prune noise.

2. Feature Scale Harmonization

Normalize binary one-hots to eliminate sex shortcut ($\Delta\|x\|^2 = 2$).

3. Boilerplate Disentangling

Decouple missing-specialty boilerplate from acute symptoms.

4. Surrogate SHAP Explainers

Fit Tree/KernelSHAP to quantify non-linear risk attributions.

Key Takeaway: AI multiplies implementation velocity; human visual auditing and domain inquiry ensure clinical validity.